



ST BENEDICT'S COLLEGE

SUBJECT	LIFE SCIENCES	DATE	16 MAY 2022
GRADE	11	MARKS	50
EXAMINER	MRS S. STEGMANN	MODERATOR	MRS B. ZAJAC
		DURATION	50 MINUTES

MARKING GUIDELINES

Question 1:

- 1.1.
- 1.1.1. A ✓
- 1.1.2. B ✓
- 1.1.3. C ✓
- 1.1.4. B ✓
- 1.1.5. C ✓ (5)
- 1.2.
- 1.2.1. Pulmonary veins ✓
- 1.2.2. Lymph ✓
- 1.2.3. Diastole/ general diastole ✓
- 1.2.4. Phagocytes ✓
- 1.2.5. Heart rate ✓ (5)
- [10]

Question 2:

- 2.1. 1- superior vena cava ✓ 2- inferior vena cava ✓ 3- pulmonary artery ✓ (3)
- 2.2. It separates the left and right side of the heart ✓ ensuring that oxygenated and deoxygenated blood do not mix. ✓ (2)
- 2.3.
- 2.3.1. Contractions arise within the muscle itself. ✓ (1)
- 2.3.2. A✓ - Sinoatrial (SA) node ✓ (2)
- 2.3.3. The SA node (A) generates contractions in the right atrium ✓
This stimulates the AV node (B) causing the atria to contract ✓
Which in turn sends impulses down the Bundle of His (C) ✓
These branch into the Purkinje fibres (D) which stimulate contractions in the ventricles. ✓ (4)
- 2.3.4. The parasympathetic nervous system/ the nervous system ✓
Impulses are sent along the parasympathetic NS to the SA node ✓
The SA node reduces the number of impulses that it sends out and HR slows down ✓ (3)
- [15]

Question 3

- 3.1. 14 ✓kPa ✓ (2)
- 3.2. The aorta receives blood directly from the left atrium, which pumps blood at the highest pressure. ✓
The aorta has a thick muscular wall to maintain the pressure of the blood. ✓ (2)
- 3.3. a. The contracting and relaxing ✓ of the muscular wall/layer ✓ of the arteries. (2)
- b. Heading ✓
Accuracy and diagram rules ✓
Any 3 labels ✓✓✓ (5)
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- 3.4. a. As blood moves from the aorta to smaller arteries and arterioles there is a decrease in blood pressure. ✓ (1)
- b. Capillaries only consist of one layer of endothelium. ✓
High blood pressure will cause the walls of the capillary to rupture/burst. ✓ (2)
- 3.5. SL valves ✓ (1)

[15]

Question 4

- 4.1. Heading ✓
Enclosed table ✓
Column headings ✓
Data ✓ (4)
- 4.2. Patient 2 (120/80) ✓ (1)
- 4.3. Ventricular systole ✓ (1)
- 4.4. Hypertension ✓ (1)
- 4.5. Controlled: weight, smoking, exercise, any relevant answer (any 1 ✓)
Uncontrolled: age, genetics, being male (any 1 ✓) (2)
- 4.6. Heart attack, aneurysm, stroke, heart failure, kidney damage, brain damage (any 1 ✓) (1)

[10]

TOTAL: 50 MARKS